Date __/__/____

Lec 30
Magic Method

Magic Method :

1. __init__ : When
   When we create an object

2. __str__ :
   When we print an object
   Eg. print(student1)

3. __add__ :
   When we add two objects.

4. __sub__ :
   When we subtract two objects.

5. __mul__ :
   When we multiply two objects

6. __truediv__ :
   We use to divide two objects

Fraction :

$$\frac{A}{B} + \frac{C}{D} = \frac{A \times D + B \times C}{BD}$$

$$\frac{A}{B} - \frac{C}{D} = \frac{AD - BC}{BD}$$

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

$$\frac{A}{B} \div \left(\frac{C}{D}\right) = \frac{A}{B} \times \frac{D}{C} = \frac{AD}{BC}$$

What is fraction ? :

Numerator
denominator

eg. $\frac{1}{2}$ ← fraction

0.5 ← decimal

Example :

$\frac{1}{2}$, $\frac{7}{11}$

Fraction1 Fraction

$$\frac{1}{2} + \frac{7}{11} = \frac{1 \times 11 + 2 \times 7}{2 \times 11} = \frac{11 + 14}{22} = \frac{25}{22}$$

Subtraction :

$$\frac{1}{2} - \frac{7}{11} = \frac{1 \times 11 - 2 \times 7}{2 \times 11} = \frac{11 - 14}{22} = \frac{-3}{22}$$

Multiplication :

$$\frac{1}{2} \times \frac{7}{11} = \frac{1 \times 7}{2 \times 11} = \frac{7}{22}$$

Division :

$$\frac{1}{2} \div \frac{7}{11} = \frac{1}{2} \times \frac{11}{7} = \frac{1 \times 11}{2 \times 7} = \frac{11}{14}$$

Creating Fraction class:

```
class Fraction :
    def __init__(self, numberator, denominator):
        self.numerator = numberator
        self.denominator = denominator

    def __str__(self):
        return "{} / {}".format(self.numerator, self.denominator)

    def __add__(self, other):
        new_num = (self.numerator * other.denominator +
                   self.denominator * other.numerator)
        new_den = (self.denominator * other.denominator)
        return Fraction(new_num, new_den)

    def __sub__(self, other):
        new_num = (self.numerator * other.denominator -
                   self.denominator * other.numerator)
        new_den = (self.denominator * other.denominator)
        return Fraction(new_num, new_den)

    def __mul__(self, other):
        new_num = (self.numerator * other.numerator)
        new_den = (self.denominator * other.denominator)
        return Fraction(new_num, new_den)

    def __truediv__(self, other):
        new_num = (self.numerator * other.denominator)
        new_den = (self.denominator * other.numerator)
        return Fraction(new_num, new_den)
```

$\frac{A}{1} + \frac{C}{7} = \frac{AxD + BxC}{BD}$

```
fraction1 = Fraction(1, 2)
fraction2 = Fraction (7,11)
print(fraction1.numerator)
print (fraction1.denominator)
print (fraction2.numerator)
print(fraction2.denominator)
print(fraction1)
print (fraction2)
print(fraction1 + fraction2)
print (fraction1 - fraction2)
print (fraction1 * fraction2)
print (fraction1 / fraction2)
```

```
> 1
  2
  7
  11
  1/2
  7/11
  25/22
  -3/22
  7/22
  11/14
```